SRM UNIVERSITY

WORKLOAD AND TIMETABLE PORTAL

Universal University Academic Workload&Timetable ERP

Requirements&Functional Documentation — Version 0.1

DOCUMENT STATUS

This document captures the requirements and examples provided so far. It is a requirements baseline, not the final architecture, database schema, API contract, UI design, or implementation plan. Additional requirements will be incorporated in later versions.

Prepared from the user-provided workload/timetable requirements document and the attached timetable reference image.

1. Purpose and Vision

The proposed application is a universal, institution-wide ERP portal for SRM University focused on academic workload allocation, workload management, curriculum allocation, and timetable generation and management.

The system is intended to replace fragmented/manual academic workload and timetable processes with a live, dynamic, centralized application that can serve numerous departments, programs, hundreds of faculty members, department coordinators/HoDs, administrators, and senior academic stakeholders.

The application must remain configurable because academic structures, workload requirements, faculty assignments, subjects, rooms, timings, and timetable rules may differ between departments, programs, semesters, academic years, and faculty members.

1.1 Core Objectives

- Provide a centralized workload and timetable platform for the whole institution.
- Allow authorized administrators to manage academic and faculty data.
- Allow faculty to submit subject preferences through a willingness-form-style workflow.
- Allow administrators to review, approve, keep pending, deny, and ultimately allocate subjects to faculty.
- Generate faculty workload information from finalized allocations.
- Simultaneously generate the Class-Wise Curriculum Allocation Matrix from the same allocation data.
- Generate conflict-free class/section timetables and individual faculty timetables using configured academic rules and constraints.
- Provide live access to workload, assigned subjects, and personal timetables.
- Preserve historical records and changes across departments, faculty, allocations, workload, and timetables.
- Provide PDF export of generated workload information.

2. Intended Users and Scope

2.1 Primary User Groups

User / Stakeholder	Expected Responsibility	Status
University-level academic stakeholders	Institution-wide visibility and oversight; exact permissions to be defined later.	Mentioned; details pending
Department Admin / Department Coordinator / HoD	Manage faculty, workload allocation, timetable generation, review/editing, and department academic data.	Core requirement
Faculty	Login, submit subject preferences, view assigned subjects, workload, and personal timetable.	Core requirement
Dean / senior academic roles	Institutional oversight is part of the broader vision; exact workflow/permissions are pending.	Mentioned; details pending

2.2 Universal Institution-Wide Scope

The ERP is not intended to be a single-department application. It must support many departments and programs under one institutional platform. Department, program type, year, semester, section, subject, faculty, room, and timetable context must therefore be selected or resolved dynamically rather than hardcoded.

3. Academic Context Selection

When an administrator enters the workload/timetable workspace, the application should provide a structured academic context selection flow.

1. Select the department, for example MCA, B.Sc., or another institution-supported department/program context.
2. Select the course/program type: Undergraduate (UG) or Postgraduate (PG).
3. Select the applicable program year, such as B.Sc. 1st/2nd/3rd Year or M.Sc. 1st/2nd Year.
4. Select the semester type: Odd Semester or Even Semester.
5. Load the subjects, sections, faculty, allocations, workload rules, and timetable configuration relevant to the selected context.

The selected context must drive the data shown and the operations performed by the administrator. The same application should be reusable across departments, UG/PG programs, academic years, semesters, and sections.

4. Faculty Account Provisioning and Access

4.1 Faculty Master Data Maintained by Admin

The administrator is expected to have faculty details including, at minimum:

- Faculty name
- Designation
- Phone number
- ERP ID
- Email ID

4.2 Faculty Login Provisioning

6. The administrator creates/provisions a login for each faculty member in the department.
7. The system uses the faculty member's institutional email address and existing ERP ID as part of the login/access process, according to the final security design.
8. The administrator should be able to send faculty access/invitation information by email to the registered email ID.
9. The faculty member should then be able to access the portal smoothly and securely.

SECURITY NOTE

The requirement currently states that faculty use their email ID and ERP ID for login. The final authentication

mechanism, password/OTP/SSO approach, credential recovery, session management, and security controls are intentionally not fixed in this requirements version.

5. Faculty Subject Preference / Willingness Workflow

The faculty portal must provide a subject-preference submission workflow similar to a willingness form. A faculty member should be able to express preferences before the administrator finalizes workload allocations.

10. Faculty logs into the portal.
11. Faculty selects the relevant department.
12. Faculty selects the relevant specialization, where applicable.
13. Faculty indicates Undergraduate, Postgraduate, or both. A faculty member may select both UG and PG subjects.
14. Faculty selects Odd Semester or Even Semester.
15. The system presents relevant subjects for the selected context.
16. Faculty selects the subjects they prefer.
17. Faculty selects/records the applicable subject category.
18. Faculty submits the preferences.
19. The submission becomes visible to the administrator in the Faculty Management area.

5.1 Admin Review of Preferences

For each faculty member and each submitted subject preference, the administrator should be able to review the request and set its status:

- Approve
- Keep Pending
- Deny

The administrator must not be restricted to the faculty member's submitted preferences when making the final allocation. The administrator should be able to offer/allocate any relevant subject that the faculty member did not select, when appropriate.

6. Workload Allocation and Generation

6.1 Admin-Controlled Allocation

The workload module gives the administrator authority to finalize academic teaching allocations. The administrator must be able to manually assign faculty roles and teaching hours for subjects.

- Assign a Main Faculty member for a subject.
- Assign an Assistant Faculty member / Incharge 2 (IN-2), where applicable.
- Allocate theory hours.

- Allocate lab/practical hours.
- Support subjects handled by more than one faculty member.
- Distinguish whether a faculty member handles theory, practical/lab, or both.
- Perform the required mathematical calculations through the allocation matrix so that total workload hours credited to each faculty member are derived from allocation data.

6.2 Finalization Flow

20. Administrator reviews faculty preference submissions.
21. Administrator reviews the status of each preference.
22. Administrator offers/allocates subjects to faculty. Allocation is not limited to subjects selected by faculty.
23. Administrator verifies that required faculty/subjects in the selected department and academic context have been appropriately allocated.
24. Administrator confirms the finalized allocations.
25. Administrator selects Generate Workload.
26. The system generates finalized workload for each faculty member in tabular form.
27. The generated workload can be exported to PDF.
28. The administrator retains the ability to access and edit workload information later.

6.3 Workload Must Be Derived from Allocation Data

The requirements indicate that the allocation matrix should perform the mathematical calculations that determine total hours for each faculty member. The implementation must therefore avoid disconnected manual copies of the same workload information. The final data model must define a clear source of truth for allocations and derive workload totals from those allocations.

IMPORTANT — NO FIXED WORKLOAD ASSUMPTION

The current requirements do not establish a universal fixed workload target such as 18 hours for every faculty member. Workload targets and policies must remain configurable and will be defined from the university's actual rules when the remaining requirements are provided.

7. Class-Wise Curriculum Allocation Matrix

When the administrator allocates faculty roles and teaching hours in the workload module, the Class-Wise Curriculum Allocation Matrix (Class-Centric Module) must be formed simultaneously from the same finalized allocation information.

- The faculty-centric workload representation answers which faculty member handles which subjects/classes and how many hours are allocated.
- The class-centric representation answers which subjects and faculty are assigned to each class/section and whether required teaching coverage is represented.
- The two representations must remain mathematically and logically consistent with the underlying allocation data.

- The exact column layout and calculation rules should follow the established workload spreadsheets and workload summary supplied by the user.

The supplied requirements describe this as a bi-directional workload tallying/reconciliation concept: the system should be able to understand the allocation from both the faculty perspective and the class/section perspective.

8. Workload Management, History and Editing

- The administrator should be able to access generated workload records at any time according to permissions.
- The administrator should be able to edit workload/allocation details after generation.
- Modifications made after generation must update relevant dependent representations consistently.
- The application must preserve history and records rather than losing prior information through silent overwrites.
- The final implementation must define how historical versions, edits, approvals, and audit information are retained.

The exact approval/locking/versioning workflow is not yet finalized in the current requirements and should be defined after the remaining ERP rules are provided.

9. Faculty Portal Requirements

- Faculty must be able to log in smoothly using the institution-approved authentication process associated with their email ID and ERP ID.
- Faculty should be able to access relevant subjects across departments without unnecessary barriers, subject to final authorization and allocation rules.
- The system should prevent or clearly manage scheduling/allocation clashes.
- Faculty should have a persistent view of the subjects they handle.
- Faculty should have a workload view showing current assigned workload.
- Faculty should have access to their personal timetable.
- The interface should be simple and user-friendly for faculty who may not be technically experienced.

10. Timetable Generation and Management

After workload/allocation data is finalized, the administrator should be able to generate timetables for relevant departments, programs, classes, and sections for both undergraduate and postgraduate contexts.

10.1 Timetable Generation Inputs

- Finalized faculty workload/teaching allocations
- Class and section information
- Subject information
- Theory versus lab/practical classification
- Faculty availability
- Room/classroom availability

- Lab availability and room suitability
- Configured working days
- Configured period timings and durations
- Configured break periods
- All active timetable constraints and rules

10.2 Admin Timetable Workflow

29. Administrator selects the relevant academic/department context.
30. Administrator provides or reviews essential data such as room availability.
31. Administrator accesses the time-setting configuration.
32. Administrator sets or confirms working days, period timings, duration of each period, break timings, and other applicable timetable settings.
33. Administrator initiates timetable generation.
34. The timetable engine generates class/section timetables.
35. The system simultaneously generates personal timetables for relevant faculty.
36. The system validates generated schedules against active constraints.
37. Administrator reviews the generated result.
38. Administrator/department coordinator can edit the timetable when authorized.

11. Timetable Constraints and Rules

The following rules are captured from the supplied requirements document. They are documented as requirements, not as an assumption that every rule is technically a hard constraint. During final timetable-engine design, each rule must be classified as hard, soft, configurable, or dependent on university policy.

11.1 Working Days

Timetables should be generated only for five days: Monday through Friday.

11.2 Faculty Workload and Availability Consistency

Faculty and class timetables must match allocated teaching hours and respect the availability represented in the data.

11.3 Break Protection

No subject or faculty teaching assignment may be scheduled during a configured break.

11.4 Prefer a Lab-Free Day Where Feasible

From Monday through Friday, the generator should allow a day with no lab session when possible under other constraints.

11.5 Faculty Free Slot Requirement

A faculty member must have at least one free slot in the weekly timetable, subject to the final definition of the applicable timetable scope.

11.6 Fully Occupied Class Timetable

Each class/section timetable should not contain free slots and should be fully occupied, subject to the configured timetable duration and break structure.

11.7 Theory Subject Daily Repetition

For theory sections, the same subject should not be repeated more than two times in one day. This does not apply to lab sessions.

11.8 First/Last Period Distribution

The same hour/subject should not be placed as the first period or last period throughout the week. Exact scope will be formalized later.

11.9 Lab Continuity

If about two lab sessions are scheduled for a subject, they may be continuous. A particular lab subject must not exceed two consecutive lab sessions.

11.10 Different Lab Subjects

If different lab subjects occur on the same day, the same consecutive-lab rule should be respected for each subject.

11.11 Theory Continuity

Theory sessions do not have to be consecutive. If the administrator wants consecutive theory sessions, this may be supported as a configurable preference.

11.12 Theory/Lab Recognition

The application must distinguish theory subjects from lab/practical subjects during timetable generation and validation.

11.13 Lab Sessions Across a Break

If two consecutive lab sessions are intended and a configured break occurs between them, the break in the middle is allowed.

11.14 Conflict-Free Generation

The timetable generator must analyze supplied data and generate feasible schedules without unresolved faculty/class/room conflicts.

11.15 Data Inconsistency and Typo Handling

The application should recognize and gracefully handle human-entered data inconsistencies and typographical variations. Exact correction/normalization policy remains to be defined.

11.16 Timetable Display Content

Every timetable slot should display subject name or suitable short form/acronym, subject code, faculty name, and allocated classroom/lab room number.

11.17 Admin-Defined Timing

When the administrator or department coordinator sets timetable timings, the generator must adhere to configured period durations, start/end times, breaks, and other timing settings.

11.18 Additional Rules

The application must support addition of new timetable rules by an authorized administrator. The final rule-definition mechanism must be designed later.

11.19 Mixed Practical Day

A class day may contain practical/lab sessions from different subjects across the day, including multiple lab subjects separated by breaks.

11.20 Maximum Daily Lab Hours per Subject

The same lab subject must not be conducted for more than two hours in a day, whether consecutive or non-consecutive.

12. Elective and Language Subject Scheduling Requirements

The requirements identify elective and language subjects as a special scheduling case. These are not ordinary independent class entries because multiple elective choices may occur simultaneously and may involve different student groups, faculty, rooms, or mixed sections.

12.1 Simultaneous Elective Scheduling

If an elective option such as Elective R is represented by multiple alternatives (for example Elective A, Elective B, and Elective C), the alternatives may need to commence at the same timetable period while being delivered in different rooms/labs by different faculty members.

12.2 Mixed-Section Elective Groups

An elective class may combine students from two or more sections. The combined student group may attend the same subject, in the same venue, at the same time, with the faculty member handling the combined group.

12.3 Language Subjects

Language subjects may follow the same simultaneous/multi-group scheduling pattern described for electives and must be handled without faculty, room, or student-group conflicts.

DESIGN DECISION PENDING

The current requirements intentionally do not force a specific implementation for elective/language scheduling. The final data model and timetable engine must explicitly represent synchronized groups, student/section membership, shared venues, and faculty assignments before implementation.

13. Practical/Lab Day Example Captured from the Requirements

A class timetable may contain a full day of practical/lab sessions belonging to different subjects. The supplied example describes a possible sequence such as:

- First hour: Java Lab
- Next two hours: Python Lab
- Break
- One hour: DCN Lab
- Last two hours: IoT Lab

Different lab subjects may occupy the same day, but the same lab subject must not exceed two hours in that day, regardless of whether those hours are consecutive or non-consecutive.

14. Timetable Presentation Requirements

The generated timetable must be presented in a clean, readable tabular structure. The attached current timetable image is the visual reference for the general style and information density, not a declaration that the exact spreadsheet formatting must be reproduced.

- Monday to Friday must be clearly represented.
- Each configured period/time slot must be clearly labeled with its timing.
- Break periods must be visibly distinguished.
- Each timetable cell should contain subject name or short form, subject code, faculty name, and classroom/lab room number.
- Class/section timetables should be available for each class and section.
- Faculty personal timetables should be available for each faculty member.
- The display must remain readable when subject or faculty names are long.
- The timetable should be suitable for web viewing and later export/printing.

15. Extracted Data from the Supplied Current Timetable Reference

The attached image shows a current timetable/workload-style sheet from SRM Institute of Science and Technology, Ramapuram, Chennai. The following details are transcribed from the visible reference image and are included only as a sample/reference dataset.

Field	Extracted Reference Value
Institution	SRM Institute of Science and Technology – Ramapuram, Chennai
Faculty / School	Faculty of Science and Humanities
Department	PG Department of Computer Applications

Programme	MCA GEN AI
Section	II MCA GEN AI A
Semester	ODD Semester 2026– 27
Reference timetable days	Monday to Friday
Visible periods	Period 1 through Period 6
Visible period timings	1: 8:00– 8:45; 2: 8:45– 9:30; 3: 9:30– 10:15; 4: 10:30– 11:15; 5: 11:15– 12:00; 6: 12:00– 12:45
Visible break	10:15– 10:30

15.1 Extracted Course / Workload Reference Table

Stream / Title	Course Code	Course Name	Faculty Member(s)	Dept	Credits	Allotted Hours
C	PCA25C08J	Data Communication Networks	Dr M. Sowmya; Mrs M. Divya	MCA	4	5
C	PCA25C09T	Cloud and Quantum Computing	Dr K. Kottisamy	MCA	4	4
D	PGI25D07J	Computer Vision in Smart Robotics	Dr S. Uma Rani; Dr K. Kottisamy	MCA		
D	PGI25D08J	Building Conversational AI for Human Resources	Dr. M. Sakthi Aswini; Mrs M. Divya	MCA	4	5
D	PGI25D09J	IoT Devices with Computer Vision Technologies	Dr. N. Krishnamoorthy; Dr S. Chithra	MCA		
G	PGI25G04T	Data Engineering and Analytics	Mrs M. Divya	MCA		
G	PGI25G05T	Distributed Data Processing Systems	Dr K. Kalaiselvi	MCA	2	2
G	PGI25G06T	Quantum Machine Learning	Dr. N. Krishnamoorthy	MCA		
C	PCA25C10L	Capstone Project	Dr M. Sowmya; Dr J. Dhilipan	MCA	4	14
P	PGI25P01L	Internship	Dr M. Sowmya	MCA	2	0
		Total Credits			**20**	**30**

Note: Blank credit/hour cells above are intentionally left blank where the supplied image does not visibly provide a value. They are not inferred or filled using outside assumptions.

15.2 Reference Timetable Layout

The current reference timetable uses rows for Monday through Friday and columns for numbered periods. It visibly separates the break between the third and fourth periods and places course/subject entries within each time slot. The ERP should support the same underlying concept while generating the layout dynamically from configured timetable settings.

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY - RAMAPURAM, CHENNAI

FACULTY OF SCIENCE AND HUMANITIES

Department: PG Department of Computer Applications — Programme: MCA GEN AI

Section: II MCA GEN AI A — Semester: ODD Semester 2026-27

Day|Session	1 (8.00 -8.45)	2 (8.45 -9.30)	3(9.30 -10.15)	BREAK (10.15 - 10.30)	4(10.30 - 11.15)	5(11.15 - 12.00)	6(12.00 - 12.45)
MON	DCN LAB - PCA25C08J		DEA - PGI25G04T DDPS - PGI25G05T QML- PCA25G06T		CSP - PCA25C10L	CQC - PCA25C09T	CSP - PCA25C10L
TUE	BCAIHR LAB -PGI25D08J IOTCV - PGI25D09J CVSR LAB - PGI25D07J		CQC - PCA25C09T		DEA - PGI25G04T DDPS - PGI25G05T QML- PCA25G06T	CSP - PCA25C10L	CSP - PCA25C10L
WED	CQC - PCA25C09T	BCAIHR -PGI25D08J IOTCV - PGI25D09J CVSR - PGI25D07J	DCN - PCA25C08J		CSP - PCA25C10L	CSP - PCA25C10L	CSP - PCA25C10L
THU	DCN - PCA25C08J	CQC - PCA25C09T	CSP - PCA25C10L		CSP - PCA25C10L	CSP - PCA25C10L	CSP - PCA25C10L
FRI	CSP - PCA25C10L	DCN - PCA25C08J	BCAIHR -PGI25D08J IOTCV LAB - PGI25D09J CVSR - PGI25D07J		BCAIHR -PGI25D08J IOTCV LAB - PGI25D09J CVSR - PGI25D07J	CSP - PCA25C10L	CSP - PCA25C10L

Career Stream Title	Course Code	Course Name	Name of the Faculty Member	Dept	Credits	Allotted Hours
C	PCA25C08J	Data Communication Networks	Dr M.Sowmya Mrs M.Divya	MCA	4	5
C	PCA25C09T	Cloud and Quantum Computing	Dr K.Kottaisamy	MCA	4	4
D	PGI25D07J	Computer Vision in Smart Robotics	Dr S Uma Rani Dr K.Kottaisamy	MCA	4	5
D	PGI25D08J	Building Conversational AI for Human Resources	Dr.M.Sakthi Asvini Mrs M.Divya	MCA		
D	PGI25D09J	IoT Devices with Computer Vision Technologies	Dr. N. Krishnamoorthy Dr S.Chithra	MCA		
G	PGI25G04T	Data Engineering and Analytics	Mrs M.Divya	MCA	2	2
G	PGI25G05T	Distributed Data Processing Systems	Dr K.Kalaiselvi	MCA		
G	PGI25G06T	Quantum Machine Learning	Dr. N. Krishnamoorthy	MCA		
C	PCA25C10L	Capstone Project	Dr M.Sowmya Dr J .Dhilipan	MCA	4	14
P	PGI25P01L	Internship	Dr M.Sowmya	MCA	2	0
Total Credits					20	30

Timetable Coordinator **HoD** **VP-Academic** **Dean (FSH)**

Figure 1. User-provided current timetable reference image.

16. Consolidated End-to-End Functional Workflow Captured So Far

1. Admin authenticates and enters the authorized workspace.

2. Admin selects department, UG/PG program type, program year, semester, and other relevant academic context.

3. System loads applicable faculty, subjects, sections, curriculum, workload settings, room data, availability, and timetable configuration.

4. Admin provisions faculty access and sends faculty login/invitation information through registered email IDs.

5. Faculty logs in and submits subject preferences through the willingness-form workflow.

6. Admin reviews each faculty preference and marks each subject as Approved, Pending, or Denied.

7. Admin reviews overall faculty preference data and finalizes subject allocations. Admin may allocate relevant subjects even if a faculty member did not request them.

8. Admin assigns teaching roles such as Main Faculty and Assistant Faculty / IN-2 and allocates theory/lab hours.

9. System calculates faculty workload from finalized allocation data.

10. System simultaneously forms/updates the Class-Wise Curriculum Allocation Matrix.

11. Admin verifies that required subjects, faculty assignments, sections, and hours are appropriately covered.

12. Admin generates finalized workload.

13. System displays workload in tabular form and makes it available for PDF export.

14. Admin supplies/reviews room availability and timetable settings.

15. Admin sets working days, period timings, durations, breaks, and active timetable rules.

16. Timetable engine generates class/section timetables and faculty personal timetables.

17. System validates generated timetables against applicable constraints.

18. Admin/department coordinator reviews and, where authorized, edits the timetable.

19. System stores resulting records and their history for future access, modification, reporting, and audit.

17. Core Data Categories Identified So Far

Data Category	Current Requirement / Purpose

Institutional structure	University, school/faculty, department, program, UG/PG type, program year, semester, section.
Faculty	Name, designation, phone number, ERP ID, email ID, department association, login/access state.
Subjects	Subject code, subject name, category/stream, UG/PG applicability, department relevance, theory/lab classification.
Curriculum	Program/semester/section subject requirements and required teaching hours.
Faculty preferences	Faculty-submitted subject preferences, selected context, subject category, review status, and administrative decision.
Allocations	Faculty-to-subject/class/section assignments, Main/Assistant/IN-2 role, theory hours, lab hours, and allocation state.
Workload	Derived faculty teaching totals and related workload representations.
Class-wise curriculum allocation	Class/section-centric representation of subjects, faculty, roles, and hours.
Rooms	Classroom/lab identity, availability, suitability, and other future constraints.
Timetable configuration	Working days, periods, timings, duration, breaks, and configurable timetable rules.
Faculty availability	Available/unavailable timetable slots or other applicable availability information.
Timetable entries	Day, period, subject, code, faculty, class/section, room/lab, and special grouping/synchronization information.
History and records	Changes, generated versions, modifications, approvals/statuses, and historical records.
Exports	Generated workload PDFs and future timetable/report exports.

18. System-Wide Qualities Required

- Universal: usable across the institution's departments and academic structures.
- Dynamic: data and academic settings should be configurable rather than hardcoded.
- Live and real-time: users should work with current application data and changes should be reflected appropriately across dependent views.
- User-friendly: suitable for both administrators and faculty.
- Secure: access and data operations must be controlled according to user roles and permissions.

- Robust: the system must handle large numbers of departments, faculty, subjects, classes, sections, and timetable combinations.
- Conflict-aware: timetable generation must not leave unresolved scheduling conflicts under the active constraints.
- Traceable: records and changes must be retained so the institution can understand what was allocated/generated and how it changed.
- Maintainable: the system must support additional rules and future requirements without requiring a complete rewrite.

19. Requirements Interpretation Policy for Future Development

39. New requirements will be documented first rather than immediately translated into code.
40. No department, program, semester, workload target, period count, timing, room count, or timetable rule will be hardcoded unless explicitly defined as a fixed institutional rule.
41. Where a requirement is ambiguous, the ambiguity will be recorded and clarified rather than silently resolved.
42. Where two requirements appear to conflict, both will be documented and the conflict will be surfaced for a later decision.
43. Hard constraints, soft constraints, preferences, and configurable rules will be classified only after the complete timetable requirements are available.
44. The final end-to-end workflow will be constructed only after the remaining ERP features have been provided.
45. Only after the workflow is approved should the production architecture, database model, API contracts, frontend structure, and implementation phases be finalized.

20. Open / Pending Requirements to Be Defined Later

- Complete institution hierarchy and exact administrative roles/permissions.
- Exact authentication method for faculty/admin accounts and whether institutional SSO is available.
- Detailed workload calculation formulas and workload targets/policies for different faculty categories.
- Exact Main Faculty / Assistant / IN-2 crediting rules, especially where theory and practical are shared.
- Exact curriculum source and required contact hours for each subject.
- Complete class/section creation and student-group representation.
- Elective and language grouping model, including synchronized sessions and mixed sections.
- Room/lab capacity, equipment, room type, and suitability rules.
- Faculty availability and leave/unavailability rules.
- Full timetable timing configuration, including whether different departments can use different schedules.
- Formal hard/soft constraint classification and timetable optimization priorities.
- How new admin-defined rules are created, activated, prioritized, and versioned.
- Timetable approval, locking, publishing, and revision workflow.
- Historical versioning and audit requirements in detail.
- Notification/email requirements beyond faculty login invitations.
- PDF/Excel/report layouts and institutional branding requirements.
- Dashboard and analytics requirements.

- Integration requirements with existing SRM ERP, identity systems, email systems, or other university platforms.
- Deployment, backup, disaster recovery, monitoring, and institutional security requirements.

21. Source Traceability

This version is based on two user-provided sources: (1) the uploaded workload/timetable requirements document, which contains the workload/timetable vision, timetable constraints, and elective/language concerns; and (2) the supplied timetable screenshot, which is used as a visual reference and sample data source.

The wording has been professionally reorganized and grammatically normalized while preserving the stated intent. Where the source did not provide a value, this document does not invent one.